



# Smoke test: Service Pro CRM

Run this once on a real Google account after building your master. ~15 minutes. It exercises every feature that can only be verified live. **The app is the primary interface; a few automations run from the sheet's ⚡ CRM menu.** Check each box; if one fails, note the step and the error and send it over.

Tip: use your **own email** as the test client so the emails come back to you.

## A. Build the master (sheet + code)

- ☐ **A1.** New Google Sheet → `Extensions` ▶ `Apps Script` . Add the code files: `Code.gs` (paste over the stub), plus new **script** files `Api.gs` , `db.gs` , `setup.gs` , and new **HTML** files `WebApp` , `Sidebar` (optional: `Tests.gs` ). **Save.** (Or `clasp push` , see `CLASP-SETUP.md` .)
- ☐ **A2.** Reload the sheet → a ⚡ **CRM** menu appears → ⚡ `CRM` ▶ `Set up / rebuild database` → approve the Google auth (Advanced → Go to project → Allow).
- ☐ **A3.** Confirm the **7 data tabs**: Clients, Jobs, Estimates, Invoices, LineItems, Services, Business Settings (plus a hidden `_meta` ), with **no leftover "Sheet1."** The dashboard lives in the **app** (Home screen), not a sheet tab.

## B. Deploy the app (see `DEPLOY.md` )

- ☐ **B1.** `Deploy` ▶ `New deployment` ▶ (gear) `Web app` → **Execute as: Me, Who has access: Only myself** → **Deploy** → **Authorize**. Copy the **Web app URL** (ends in `/exec` ).
- ☐ **B2.** Open the URL → the **app loads on the Home screen** (KPI tiles, "Follow-ups due", "This week's jobs"). (If you get a Drive "unable to open the file" error, you have multiple Google accounts, so open it in an Incognito window signed into just the owner account. ⚡ `CRM` ▶ 📄 `Open the app (get Link)` shows the URL anytime.)

## C. Settings & services (in the app)

- ☐ **C1.** App → **More** ▶ **Settings** → fill **Business name**, **Owner email** (your address), phone, **Currency**, **Sales tax %** (e.g. 8.25), **Google review link** (any URL), payment instructions, and a **Payment link** (any URL). Set **Invoice due (days to pay)** (e.g. 7 , or 0 for "Due upon receipt"). Under **Document numbering**, optionally set a custom **Next quote/estimate #** (e.g. Q-2001 ) and **Next invoice #** (e.g. INV-5001 ) → **Save settings**.
- ☐ **C2.** Edit the **"Your services"** box (one per line) → **Save services**. The new service now appears in the app's Service dropdowns.

## D. Contacts (leads + clients, in the app)

- ☐ **D1. Clients** ▶ + (or Home ▶ + **Lead**) → add a client with **name, phone, email (= your own), and address** → **Save**. It appears in the **Clients** list (tagged **Lead**) and in the **Clients** sheet tab, with a **Next Follow-up** date set automatically.
- ☐ **D2.** Tap the contact → **Call / Text** links show, address + service show, **New / Quoted / Declined / Lost** status buttons work, follow-up reschedule works, and **Create estimate** opens the builder pre-filled. After they have a quote/invoice/upcoming job, the detail shows an **Active** section (tap a quote/invoice → opens it in Billing).
- ☐ **D3. Add a second contact re-using the same email or phone** → a  **Possible duplicate** prompt lists the match; **Cancel** stops it, **Add anyway** still adds it (warn, not block).
- ☐ **D4.** Toggle the **Name / Last job** sort. A contact who has a logged job shows their **last job date** (green) instead of a status; sorting by **Last job** puts the most-recently-served first.

## E. Quote → Approve → Invoice → Paid: the core flow (in the app)

- ☐ **E1. Billing** ▶ + **New estimate** → client = your test client (email = you) → add **2 line items** (Service + Qty + Price) → watch the **running Total** update (with tax) → **Create & send**.
- ☐ **E2.** The estimate appears under **Billing** ▶ **Quotes and** in the  Estimates tab; you receive an **ESTIMATE PDF**, itemized, with **subtotal + tax + total** and "Valid until". (If you set a custom start in C1, confirm the estimate # matches it (e.g. **Q-2001**), and the invoice created at E3 uses your invoice start.)
- ☐ **E3.** Open that estimate → **Approve** → **create invoice**. Confirm: estimate → **Accepted** (it moves to the collapsed **Closed quotes** group); a  **Invoice draft** appears under **Outstanding invoices** with the **line items copied**; and a  **Job (Scheduled)** was created.
- ☐ **E4. Billing** ▶ **Outstanding invoices** → open the draft → (optionally **Edit line items**) → **Send invoice** → you receive an **INVOICE PDF** (itemized, tax, a **Pay now** button) and status flips to **Sent**. Confirm the **Due** = send date + **your terms** (e.g. 7 days), or reads "**Due upon receipt**" if you set **0** in C1.
- ☐ **E5. Mark paid** → status **Paid**; the invoice drops out of **Outstanding** into the collapsed **Paid** group (tap it to expand). Go to **Home** → **Revenue** and **Lifetime** update.
- ☐ **E6.** Re-open that invoice and **Send** again → the **total is unchanged** (tax is not applied twice). 
- ☐ **E7.** Open any estimate → **Mark accepted** moves it to **Closed quotes** (keeps the record); or **Delete quote** removes it (with a confirm). Open an invoice → **Delete invoice** works the same. Use these to clear stuck/duplicate docs from old testing.
- ☐ **E8.** Note your **Home** ▶ **Lifetime** figure. **Settings** ▶  **Archive old paid & closed docs** → confirm the toast counts; **Billing** now shows fewer Paid/Closed rows, and new  **Archived...** tabs hold them. Re-check **Home** ▶ **Lifetime**: it's **unchanged** (archived paid revenue is carried forward). Run archive again → it reports **nothing** to archive (idempotent).

## F. Jobs & Contacts (in the app)

- ☐ **F1. Jobs** ▶ + **New job** → add a job for the client (date, price) → appears in the app **Jobs** list and the  Jobs tab.
- ☐ **F2.** Tap a job → **Reschedule** to a new date (it moves in the list), set **Status = Done** and **Mark paid**. From **Home**, tapping a "this week's jobs" row opens that **same job detail**.
- ☐ **F3. Contacts** → your test client now shows a green **last job date** tag; open them → **Last job ...** · **N jobs** · **Total spent** = the sum of their paid jobs.

## G. Automations & emails (from the sheet's ⚡ CRM menu)

These run server-side / on triggers, not in the app.

- ☐ **G1.** ⚡ CRM ▶ Automations ▶ Send review requests → you (as the test client) get the ★ **review-request email**; the job's **Review Sent? = Yes**. Run again → **nothing sent** (no duplicate).
- ☐ **G2.** Make sure a lead's follow-up is **due today**, then ⚡ CRM ▶ Automations ▶ Email me today's follow-ups → you get the 🔔 **digest** (with a tap-to-text link). *(Also a one-click in the desktop Quick Actions panel.)*
- ☐ **G3.** Add a job dated **tomorrow** for your client → ⚡ CRM ▶ Automations ▶ Send tomorrow's appointment reminders → you get the  **reminder**; **Reminder Sent? = Yes**.
- ☐ **G4.** Set an invoice's **Due Date = yesterday**, Status = **Sent** → ⚡ CRM ▶ Automations ▶ Flag overdue invoices now → it flips to **Overdue**.
- ☐ **G5.** ⚡ CRM ▶ Automations ▶ Turn ON daily autopilot → confirmation. Apps Script ▶ Triggers shows **4 time-based triggers**. Then **Turn OFF daily autopilot** → all removed.

## H. Recurring jobs

- ☐ **H1.** In  Jobs, set a job **Repeat = Weekly** and **Status = Done**.
- ☐ **H2.** ⚡ CRM ▶ Automations ▶ Roll forward finished recurring jobs → a **new Scheduled job** appears **7 days later** for the same client; the original's **Rolled? = Yes**.
- ☐ **H3.** Run it again → **no duplicate** (idempotent).

## I. Google Calendar (now automatic, in the app)

Calendar sync is no longer a manual menu click. Scheduling a job **in the app** creates the event automatically (client address as location), and rescheduling/cancelling updates or removes it. This is covered in full by **Section O** below and by `runIntegrationSmoke` ( `Tests.gs` ).

- ☐ **I1.** In the app, add a job with a time like `2pm` (Calendar sync = On in Settings) → the event appears on your Google Calendar at 2pm. See Section O for the full lifecycle checks.

## J. Mobile lead-capture Form

- ☐ **J1.** ⚡ CRM ▶ Create mobile lead-capture form → authorize → an alert shows the **form URL** (saved in ⚙ Settings); a **"Form Responses"** tab appears.
- ☐ **J2.** Submit a test lead via that form (on your phone) → it appears in 🎯 **Leads** (New + follow-up date) **and** in the app's **Leads** list. (Security: try submitting a name like `=1+1` : it should stay literal text, not become a formula.)

## K. Safety guard

- ☐ **K1.** With data now in the sheet, run ⚡ CRM ▶ Set up / rebuild database again → a **warning** asks before erasing → click **No** → nothing is wiped.

## L. On your phone

- ☐ **L1.** Open the app URL on your phone → **Add to Home Screen** → run the **lead** → **quote** → **approve** → **invoice** → **paid** flow again. Everything works and syncs back to the sheet.

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## If something fails

Note the step (e.g. "E4: invoice PDF didn't generate") and the exact error from Extensions ▶ Apps Script ▶ Executions . The most environment-sensitive steps to check first: **B2** (deploy / multi-account), **E2 & E4** (HTML→PDF invoices/estimates), and **A3** (chart).

## After it passes

You have a verified master. Per sale: File ▶ Make a copy → share the master link (Anyone with link ▶ Viewer) → that link goes in the delivery PDF. **Each buyer deploys their own copy** (their own app URL); the deploy step is theirs, done once.

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## M. Local-first speed + optimistic writes (2026-07-17 build)

1. Open **More** ▶ **Business Settings**, scroll to the bottom: the ⌚ readout should show a low **boot** time and **0 api calls** right after opening. Navigate Home→Clients→Jobs→Billing, open a client, open an invoice, open the price book: the api-call count should **stay at 0** (all reads are local). It only ticks up when you *write*.
2. **The race:** add a new client, then IMMEDIATELY (before the "✓ Saved" badge) create an estimate for that client, then Approve → invoice + job, then Mark paid. Every step should register instantly and NONE should say "pick a valid client." Reload: all records are really there with real IDs.

3. **Reverse sync:** hand-edit a client's name in the Google Sheet, then reload the app → the edit appears (the stale flag rebuilt the JSON). Same after a lead comes in via the mobile form.
4. Tap  any time → one `apiRefresh` call, data refreshes from Sheets.

## N. Client-assigned numbers + instant actions (build g)

1. New client → shows a real `CL-` id immediately; open it and Edit works right away (no dead button).
2. New estimate → real `EST-` number immediately; Approve it right away → real `INV-` number, no "estimate not found" / "pick a valid client". Numbers never change while a screen is open.
3. From **Home**, mark a week-job Done → it leaves the Home list immediately (no ~30s wait).
4. Archive a just-created client/job → removes instantly, never "finishing save, try again."

## O. Scheduling time + Google Calendar sync + Maps directions (build j / k): GOOGLE INTEGRATION

These touch **real Google Calendar**: the automated Node suites ( `tests/*.js` ) mock the backend and **cannot** cover them, so run these by hand (or run `runIntegrationSmoke` in the editor, see below). Make sure a test client has a real **Address** first (so the event gets a location).

 **TIMEZONE: check this first.** Calendar event times use the Apps Script project's timezone. The master ships as **America/Phoenix (Arizona)**. If you (or a buyer) are in a different zone, set it in **Extensions > Apps Script > Project Settings > Time zone** to match your Google Calendar; otherwise a "2pm" job lands on the calendar at the wrong hour. Verify: schedule a `2pm` job and confirm the event shows **2pm** on your calendar, not shifted.

- ☐ **O1. Toggle exists + default on.** **More > Business Settings > Scheduling** shows **Google Calendar sync = On** and a **Default job length (hours)** field.
- ☐ **O2. First job → real event + permission.** With sync On, **Jobs > + New job** for the test client, date = a few days out, **Time = 2pm** → save. The FIRST time, Google prompts to authorize **Calendar**, approve it. Open your **Google Calendar**: an event appears on that date at **2pm**, titled with the **client name then the service**, with the client's **address as the location** (tap it → Maps).
- ☐ **O3. No time → all-day.** New job with the **Time left blank** → an **all-day** event appears.
- ☐ **O4. Reschedule updates the SAME event.** Open the O2 job → **Reschedule** to a new date/time → the calendar event **moves** to the new slot. There is **still only one** event (no duplicate).
- ☐ **O5. Cancel / archive removes it.** Set the O2 job **Status = Cancelled** (or Archive it) → the event **disappears** from Google Calendar.
- ☐ **O6. Approve carries the time. Billing** → open a Draft estimate → **Approve** → **invoice + job** → the popup asks for **Job date AND Time**; set `10am` → the created job shows `10am`, and (sync on) a **10am** calendar event appears for it.

- ☐ **O7. Toggle OFF is respected.** Settings ▶ Scheduling → **Calendar sync = Off**, Save. Create a new job → **no** calendar event is created. Turn it back **On**.
- ☐ **O8. Maps directions.** Open the test client → the **address is a link** and there's a 🗺️ **Directions** button → tapping either opens **Google Maps directions** to that address (the Maps app on a phone). From a job, tap the client first, then Directions.
- ☐ **O9. Pick-client job creation (regression).** Jobs ▶ + **New job** with **no client preselected** → tap **Pick client...** → choose a client → the form **comes back with that client set** (it must NOT vanish) → add service + date → **Add job** → the job is created.
- ☐ **O10. No field overlap.** On a phone-width screen, the **Date** and **Time** inputs on the job form, the reschedule row, and the approve popup **do not overlap** (they sit side by side).

### Automated version of Section O (optional, one click)

Tests.gs ships a `runIntegrationSmoke()` function. In **Extensions ▶ Apps Script**, pick `runIntegrationSmoke` from the function dropdown and **Run**. It creates a tagged `__SMOKE__` client + job, drives create → verify event → reschedule → cancel against your **real** calendar, then deletes the event and hard-deletes its own test rows (leaves no residue). Read the result in the **Execution log** (or the popup). This is the closest thing to "run the whole Google integration for me": the one step only you can do is approve the Calendar permission the first time. *(Dev/QA file; you can leave it out of the buyer-facing master; it's inert unless explicitly run.)*